

CSCI-3171 Network Computing

Assignment 2 – Winter 2026

Submission Instructions:

Submit **one PDF file** for your assignment on Brightspace under the Assignment 2 portal. Please ensure the following components are included:

1. Provide comprehensive steps for any computations, along with explanations of your methodology. If you made assumptions, clearly explain, and justify them. Just writing the final answer will result in points being deducted.
2. Ensure that your full name and banner ID are clearly mentioned in the report.

Late submission: Refer to the syllabus on Brightspace for a complete description of the late submission policy.

Problem 1

Suppose an HTTP client makes a request to the `www.example.com` web server. The client has never requested a given base object, nor has it communicated recently with the `www.example.com` server. You can assume, however, that the client host knows the IP address of `www.example.com`.

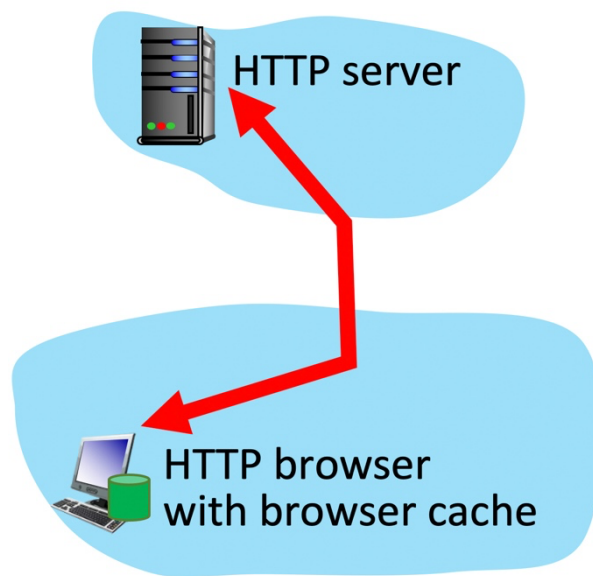
- Q1.** How many round-trip times (RTTs) are needed from when the client first makes the request to when the base page is completely downloaded, assuming the time needed by the server to transmit the base page into the server's link is equal to $1/2$ RTT and that the time needed to transmit the HTTP GET into the client's link is zero? (You should consider any TCP setup time required before the HTTP GET is sent by the client, the time needed for the server to transmit the requested object, and any propagation delays not accounted for in these amounts of time.) (1 point calculation, 1 point answer)

Suppose now that after downloading the base file, the browser encounters a jpeg object in the base html file that is stored on `www.example.com`, and therefore makes another GET request to `www.example.com` for that referenced jpeg object.

- Q2.** How many round-trip times (RTTs) are needed **from when the client first makes the request to when the base page and the jpeg file are completely downloaded**, assuming the time needed by the server to transmit the base file, or the jpeg file into the server's link is (each) equal to $1/2$ RTT and that the time needed to transmit the HTTP GET into the client's link is zero? You should assume that persistent HTTP 1.1 is being used. (You should consider any TCP setup time required before an HTTP GET is sent by the client, the time needed for the server to transmit the requested object, and any propagation delays not accounted for in these amounts of time.) (1 point calculation, 1 point answer)

Problem 2

Consider the following scenario, in which a Web browser (lower) connects to a web server (above). There is no web cache in this question (so make sure you understand the difference between a browser cache and a web cache). Assume that the total Round Trip Time (RTT) propagation and queueing delay between the browser and web server is 200 msec.



Suppose that the browser makes an HTTP request to the server for an HTTP object, that the browser has a copy of that object in its cache, but that copy may or may not be up to date in its cache (and so the browser requests the object using the HTTP If-Modified-Since header field). Assume that the time taken to transmit an HTTP reply (by the sender into the TCP connection) that contains a requested object in the body of an HTTP reply is 50 msec, but that the time taken to transmit a HTTP reply with a "304 Not Modified" reply code without an included object is negligible (e.g., zero). You can assume that a TCP connection has already been set up, so do not include that delay in your answer below.

- Q3.** What is the time from when the browser issues the initial HTTP GET request until it can display the requested object in the case that the browser does not have the requested object in its browser cache? (1 point calculation, 1 point answer)
- Q4.** What is the time from when the browser issues the initial HTTP GET request until it can display the requested object in the case that the browser **does** have the requested object in its browser cache? (1 point calculation, 1 point answer)

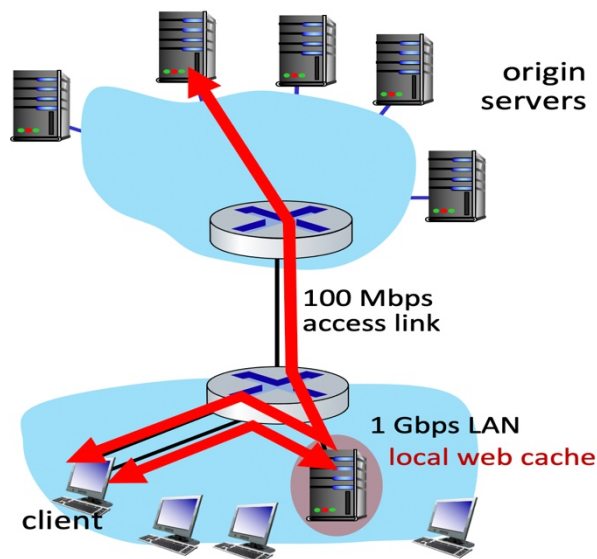
Suppose now that 90% of the HTTP requests are for objects that are up to date in the browser's cache, and so 10% are cache "misses" and will require the up-to-date object to be downloaded and browser-cached.

- Q5.** What is the average time (over all requests that the browser makes to this site) from when the browser issues the initial HTTP GET request until it can display the requested object? (1 point calculation, 1 point answer)

Problem 3

Consider the following scenario, in which a Web browser (lower) connects to a web server (above). There is also a local web cache in the browser's access network. In this question, we will ignore browser caching (so make sure you understand the difference between a browser cache and a web cache). Assume that the total Round Trip Time propagation, queueing and object transmission delay between the browser and

web server (and including TCP setup time) is 250 msec; if the object is retrieved from the local web cache, this delay is only 10 msec.



Suppose that 80 percent of the time when the browser makes a request, the requested object is found in the local web cache.

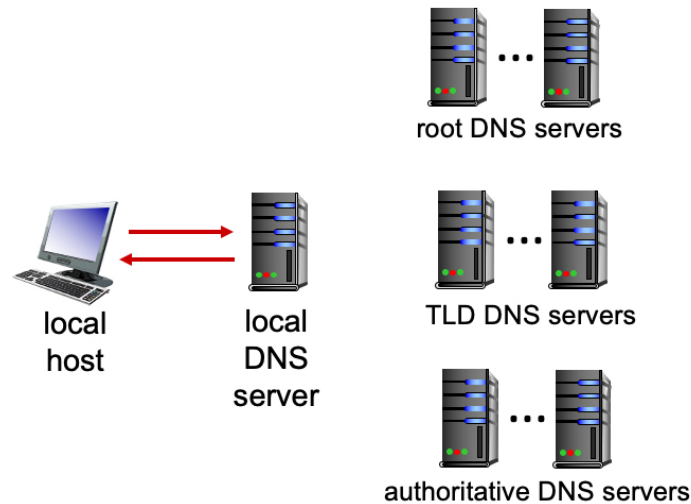
Q6. What is the average time (over all requests that the browser makes to this site) from when the browser issues the initial HTTP GET request until it can display the requested object? (1 point calculation, 1 point answer)

In the following question, we want to focus on the utilization of the 100 Mbps access link between the two networks. Suppose that each requested object is 10Mbits, and that 9 HTTP requests per second are being made to origin servers from the clients in the access network. Note that 80% of the requested objects by the client are found in the local web cache.

Q7. What is the utilization (ratio of throughput to transmission rate) of the access link? (1 point calculation, 1 point answer)

Problem 4

Suppose that the local DNS server caches all information coming in from all root, TLD, and authoritative DNS servers for 20 time units. (Thus, for example, when a root server returns the name and address of a TLD server for .ca, the cache remembers that this is the TLD server to use to resolve a .ca name). Assume also that the local cache is initially empty, that iterative DNS queries are always used, that DNS requests are just for name-to-IP-address translation, that 1 time unit is needed for each server-to-server or host-to-server (one way) request or response, and that there is only one authoritative name server (each) for any .ca domain.



We consider a PC named local host on a university network in France. The DNS server configured for this network is named local DNS server. Consider the following DNS requests, made by the local host in France at the given times:

1. $t=0$, the local host requests that the name dal.ca be resolved to an IP address.
2. $t=1$, the local host requests that the name icann.org be resolved to an IP address.
3. $t=5$, the local host requests that the name concordia.ca be resolved to an IP address. (Hint: be careful!)
4. $t=10$, the local host *again* requests that the name dal.ca be resolved to an IP address.
5. $t=12$, the local host requests that the name utoronto.ca be resolved to an IP address.
6. $t=30$, the local host *again* requests that the name dal.ca be resolved to an IP address. (Hint: be careful!)

Q8. Calculate the duration (in time units) required for each of the six requests to be resolved. Provide explanations for your results. (6 points calculation, 6 points answer)